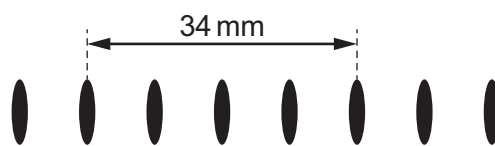


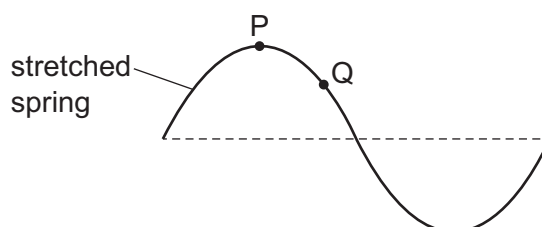
- 28 A student carries out a double-slit experiment using a laser emitting red light of wavelength of 680 nm. The light is incident normally on a double slit.

The diagram shows part of the pattern of bright fringes visible on a screen at a distance of 2.4 m from the slits. The distance across five bright fringes is measured as 34 mm.



What is the slit separation?

- A 8.5×10^{-3} m B 2.4×10^{-4} m C 1.9×10^{-4} m D 4.8×10^{-5} m
- 29 The diagram shows a stationary wave on a stretched spring at an instant in time.



Two particles on the spring, P and Q, are shown.

Which statement about the vibrations of P and Q is correct?

- A They have different frequencies.
 B They have the same amplitudes.
 C They have different periods.
 D They are always in phase.
- 30 Green laser light passes through a diffraction grating and forms an interference pattern.

The diffraction grating contains 400 lines per mm.

The wavelength of the laser light is 550 nm.

What is the highest order diffraction maximum produced by the grating?

- A 4 B 5 C 8 D 9
- 31 There is a potential difference V across a resistor of resistance R . The current in the resistor is I .

Which equation gives the power P dissipated by the resistor?

- A $P = V^2 I$ B $P = \frac{V^2}{I}$ C $P = V^2 R$ D $P = \frac{V^2}{R}$